Run 224489: Wiring Fix Confirmed, First Plastic-MIP Calibration from LIQ Triples, and HV Scan 2

n_TOF EAR2 X17 — first run with the liquid scintillators (July 17, 2026)

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analysis and slides prepared with Claude (Anthropic AI assistant)

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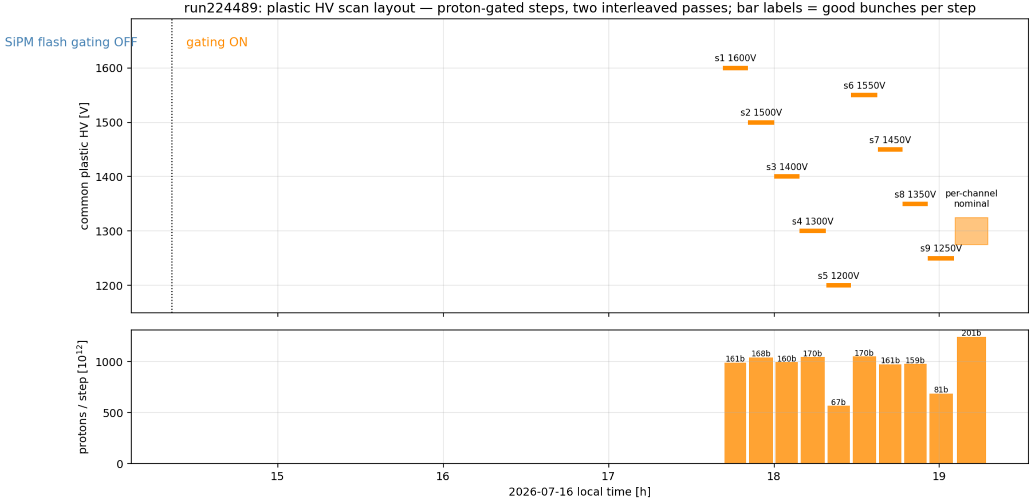
Run 224489: what changed since the last scan (224466)

- **Cabling fix** on the crossed A/D channels.
- **FIFO**: plastics on a linear fan-in/fan-out instead of the BNC-T split ($\sim$16 ns extra delay, naive 2$\times$ amplitude).
- **Liquid scintillators** (LIQA–D, 1 ch/arm) in the readout for the first time: 5.3 M (C) – 21.6 M (B) hits.
- SiPM flash gating **ON all run** (no boundary).
- Same 9-step plastic HV ladder, 1200–1600 V in 50 V merged grid, + post-scan nominal-HV window.

- 1750 bunches (817 dedicated), new PSA with LIQ pulse-shape fitting.
- QA clean: **no wall outage** (0/1750; 224404 lost 46%), γ-flash 11.62 μs on all arms, all 44 channels alive.

Scan timeline: bunches → HV steps via the CAEN log

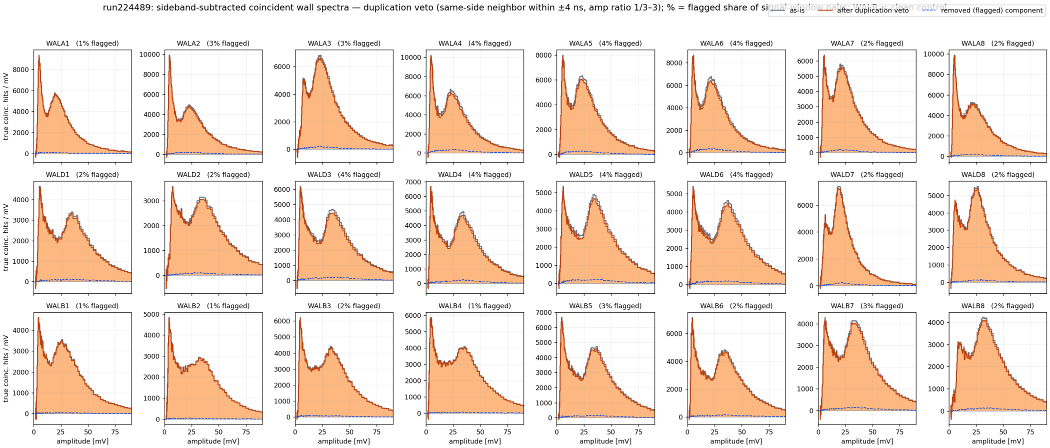


Result 1 — the cross-wiring fix is **confirmed**; A/D closed

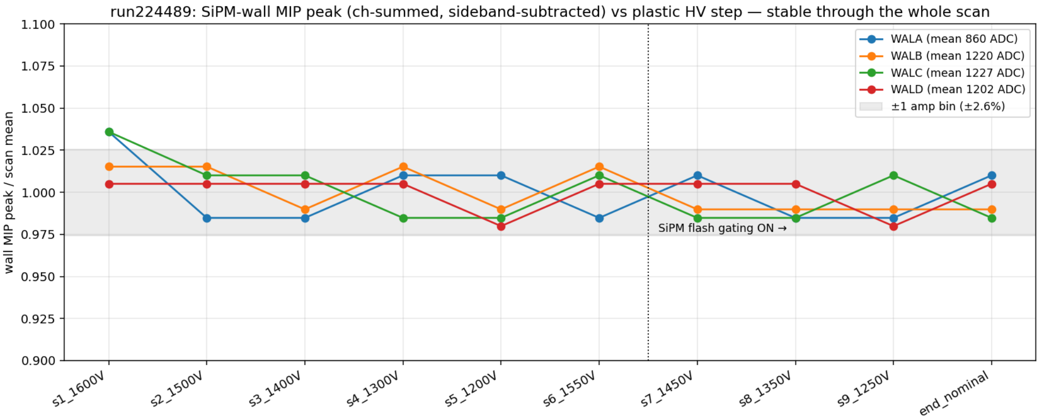
- Duplication band (± 4 ns equal-amplitude partner, one group over): flagged fractions were 23–31% on the crossed A/D channels in 224460 $\rightarrow$ now $\leq 4\%$ everywhere $=$ B-arm control level.
- Wall–plastic excess now **equal on all four arms** (1.71–1.81 M within ± 10 ns) — was 6–8$\times$ deficient on A/D.
- A/D coincident wall spectra show clean MIP bumps with **no veto**.

The standing A/D coincidence deficit was the cabling. Closed.

A/D duplication autopsy: before-and-after

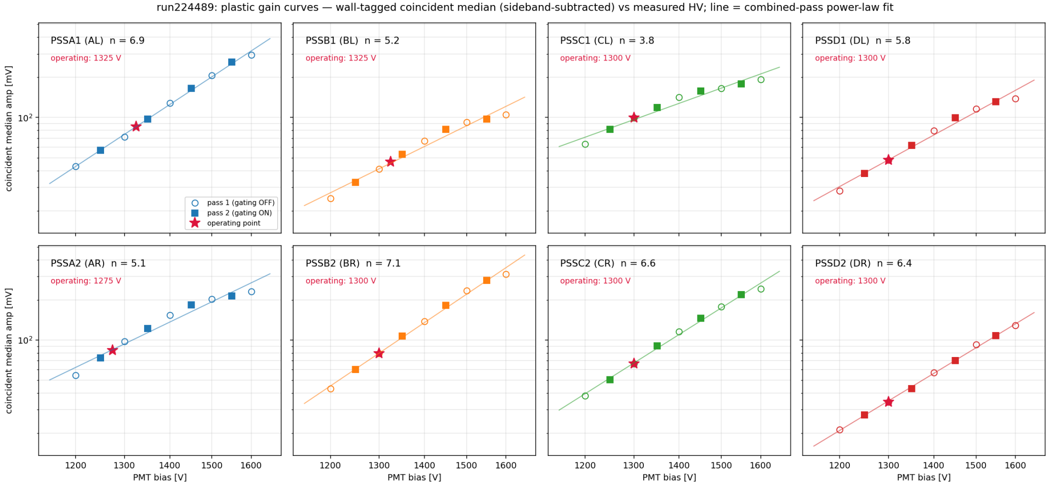


HV scan: wall MIP again immune to plastic HV (now incl. arm A)

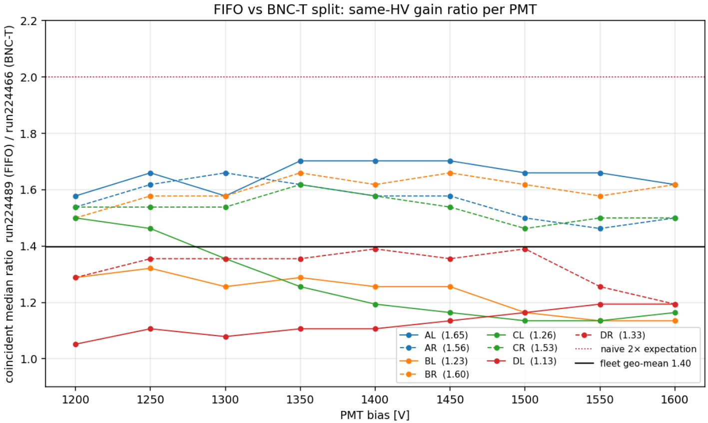


±1 bin (±2.6%) over a 3× tag-rate change ⇒ wall energy scale independent of plastic operating point.

Per-PMT gain power laws $G \propto V^n$, $n = 3.8$–7.1



Result 2 — the FIFO gain ratio is **1.4**, not 2



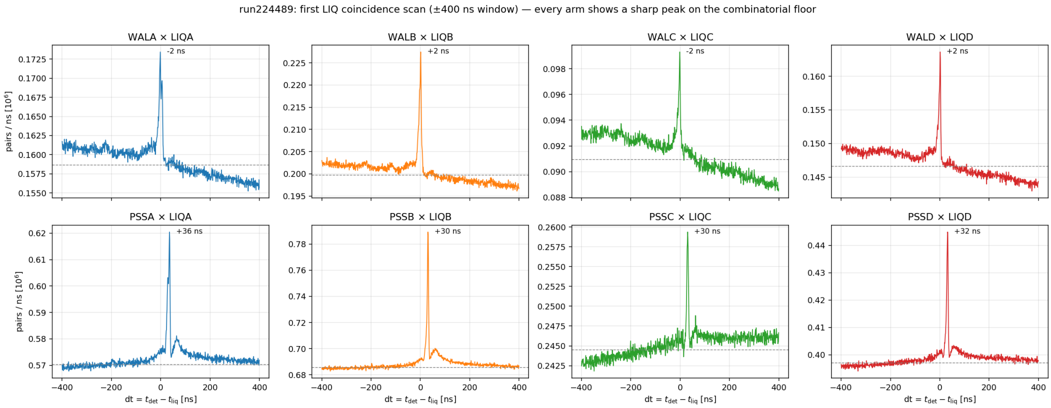
Same-HV coincident medians, 224489/224466: fleet geo-mean ×1.40, per-PMT 1.13 (DL) – 1.65 (AL). Use the measured per-PMT ratios, not a global 2.

New HV equalization (supersedes 224466 table)

PMT	med [mV]	V_{sugg}	ΔV
AL	85.4	1271	−54
AR	83.6	1211	−64
BL	46.7	1409	+84
BR	79.2	1262	−38
CL	99.8	1158	−142
CR	66.5	1293	−7
DL	47.8	1368	+68
DR	34.4	1433	+133

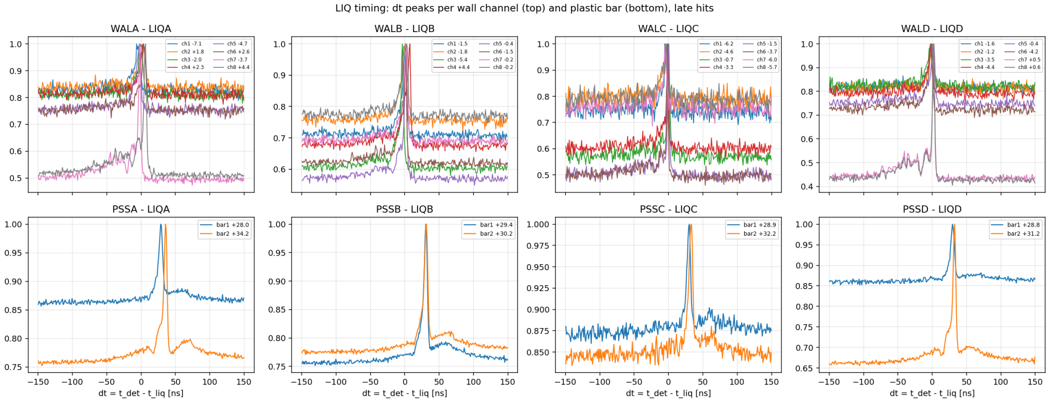
- Target: fleet geo-mean coincident median (64.2 mV); spread now 2.9×.
- At equalized gains: common threshold with 99% tagged efficiency on every PMT down to **4.9 mV** (95%: 7.1 mV).
- Exported to DAQ repo (`calibrations/pss/`, `pss_trigger/`).

Result 3 — the liquids are timed in (first time)

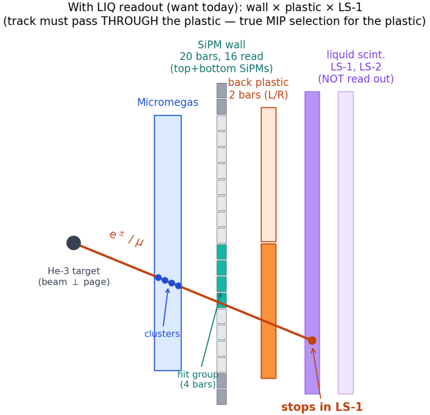


LIQ leads the plastic by 29.5–32.8 ns, within ±3.3 ns of the wall; per-channel offsets (rms 3.2–4.5 ns) → `calib/liq_offsets_run224489.json`.

LIQ timing per wall channel / plastic bar

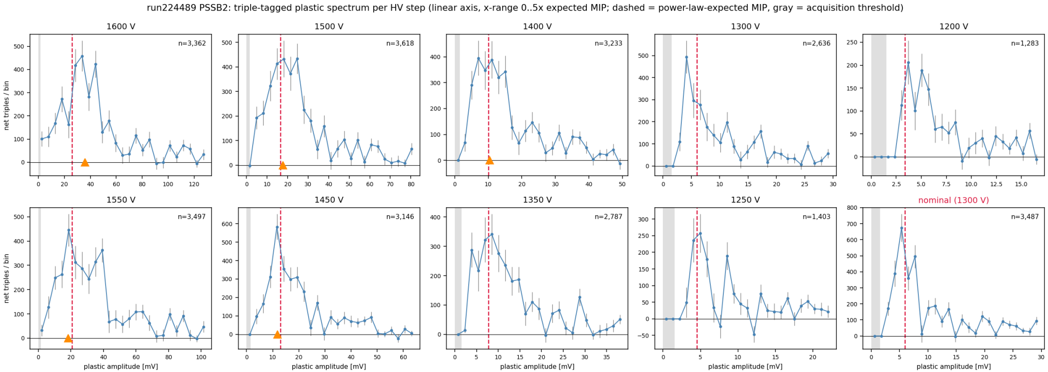


The triple telescope: why LIQ unlocks the *plastic* MIP



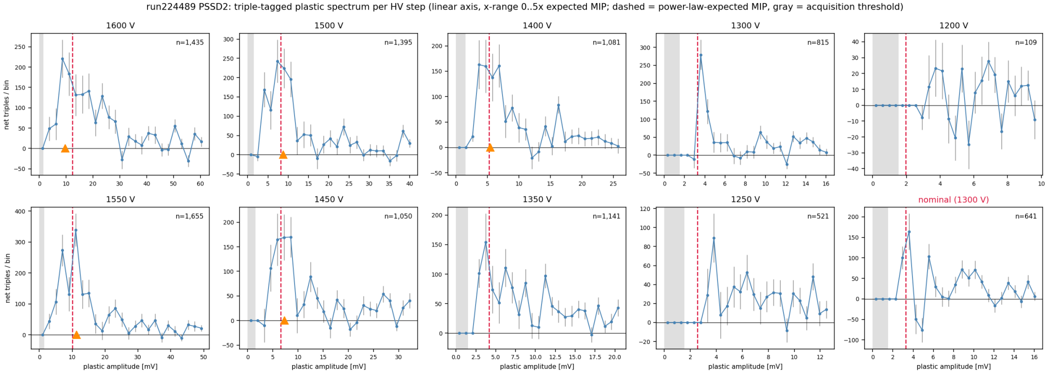
Stack: MM → wall → plastic → LS. Wall×plastic tags wall MIPs only; adding a LIQ hit *behind* the plastic selects through-goers ⇒ plastic spectrum in triples = plastic MIP candidates. Double sideband subtraction in both dt's.

Result 4 — the plastic MIP, found (PSSB2, one HV step per panel)



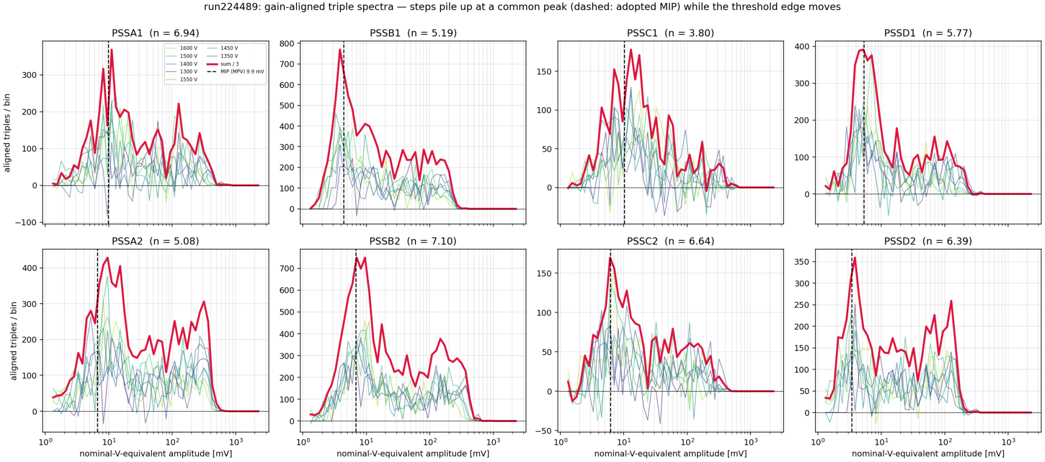
Linear amplitude axis, coarse bins, propagated Poisson errors. A single significant peak tracks the power-law expectation (dashed) panel after panel while the acquisition threshold (gray) stays put — a threshold artifact cannot do this. Orange: per-step fitted MPV. All 8 PMTs in `figures/19_triples/steps_linear/`.

Weakest PMT (PSSD2): clear at high V, rides threshold at nominal



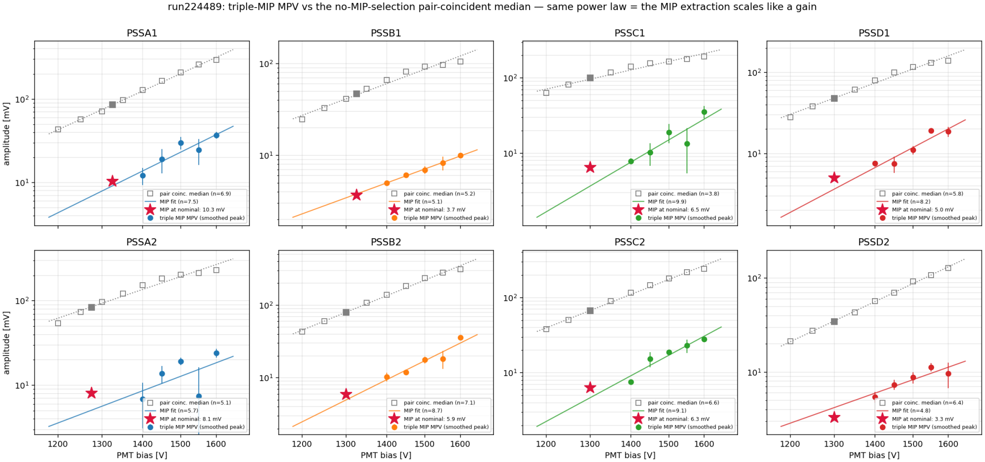
Peak clear of threshold only for $V \geq 1400$ V $\Rightarrow$ only those steps enter the fit; at nominal HV the DR MIP rides the threshold — this arm cannot trigger on plastic MIPs until equalized.

Cross-check: gain-aligned stack



Each step's spectrum scaled to its nominal-V equivalent ($\times(V_{\rm nom}/V)^n$ = translation in log amp): the steps pile up at a common peak and the sum (red) peaks at the adopted MPV (dashed) on every PMT. (Log-amp representation — peak positions here sit above the linear MPV.)

Trust check: MIP MPV vs the no-MIP-selection coincident median



Gray squares: pair-coincident median (no MIP selection, ∼100× stats). Colored: per-step triple-MIP MPVs (bootstrap errors) + own power-law fit. **Same slope, factor ∼10 below** — the MIP extraction scales like a gain; noise would not. Star: adopted aligned-sum MPV at nominal.

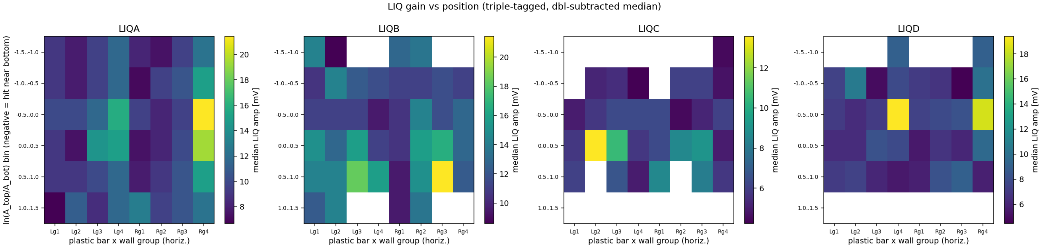
First plastic-MIP calibration table (linear-space MPV)

PMT	MPV [mV]	spread	mV/MeV
AL	10.3 ± 1.5	16%	2.05
AR	8.1 ± 1.2	42%*	1.60
BL	3.7 ± 0.5	2%	0.73
BR	5.9 ± 0.5	16%	1.18
CL	6.5 ± 1.6	38%*	1.28
CR	6.3 ± 0.3	18%	1.25
DL	5.0 ± 0.6	18%	0.99
DR	3.3 ± 0.3	14%	0.65

*AR/CL genuinely less certain (per-step scatter). Energy scale: 5.05 MeV in 2.5 cm PVT (normal incidence).

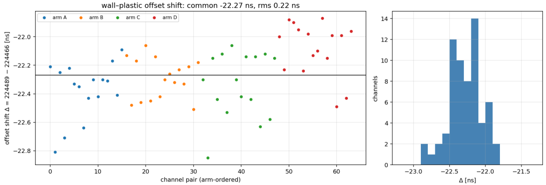
- MPV = smoothed peak of the gain-aligned summed linear spectrum ($V \geq 1400$ V steps); $\pm$ = Poisson bootstrap; spread = per-step transport scatter (the honest uncertainty on weak PMTs).
- Linear-space MPV convention: first-pass log-binned modes were 15–35% high (mode of $dN/d\log A \neq$ MPV of dN/dA).
- Validated against the no-MIP-selection median (previous slide): same power law, factor $\sim$10 below.
- DR (3.3 mV) and BL (3.7 mV) MPVs sit below the 4.9 mV trigger threshold $\Rightarrow$ equalize before trusting plastic triggers on those arms

Result 5 — LIQ gain-vs-position gradient is **real**

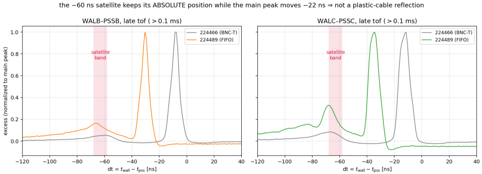


Triple-tagged track position: horizontal = wall group × bar, vertical = wall top/bottom asymmetry. 1.5–2× gradient toward the PMT; axis flips with surveyed orientation: horizontal on A/D (PMT toward group 4), vertical on B/C (PMT up) — matches the as-built Geant geometry.

Timing: one clean common delay; the satellite misbehaves



All 64 wall–plastic offsets shifted by a **common** -22.27 **ns (rms 0.22 ns)**.

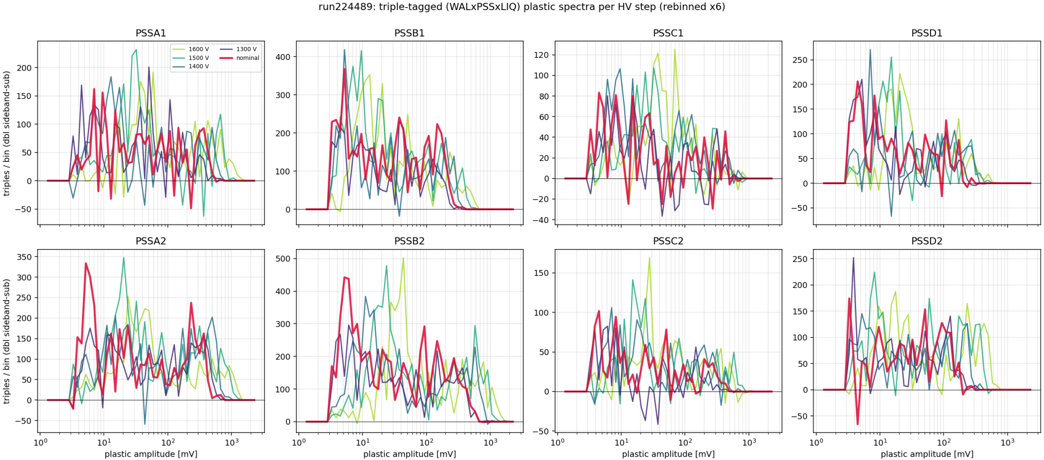


The -60 ns satellite kept its *absolute* position (main peak moved -22 ns) and grew to 19–34% of the peak $\Rightarrow$ not a plastic-cable reflection; injected at/after the FIFO. Scope check recommended. (Outside all analysis windows.)

Summary

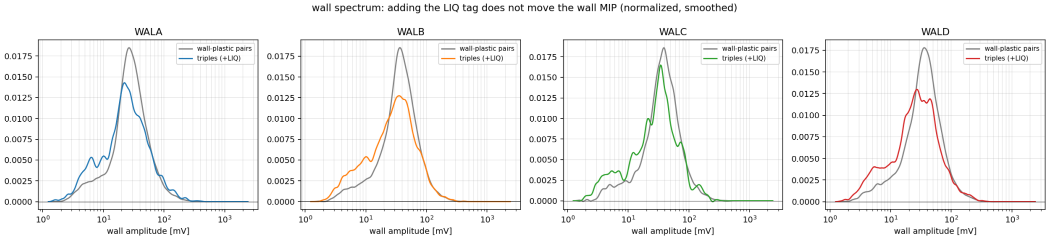
1. **A/D cross-wiring fixed and confirmed** — the old A/D deficit is fully explained; wall MIP calibration now covers all four arms.
2. **First plastic MIP calibration**: 0.65–2.05 mV/MeV at nominal HV via WAL×PSS×LIQ triples.
3. **FIFO ratio measured**: $\times 1.40$ fleet (1.13–1.65 per PMT), not $\times 2$.
4. **New HV equalization** ready (DAQ repo); apply it — DR/BL cannot trigger on plastic MIPs at 4.9 mV until then.
5. **Liquids timed in**; LIQ position-gain gradient (1.5–2×) confirmed and mapped; C/D LIQ gains look low — review LIQ HV.
6. **Scope-check the FIFO** for the -60 ns satellite.

Backup: per-step triple spectra (low stats per step)

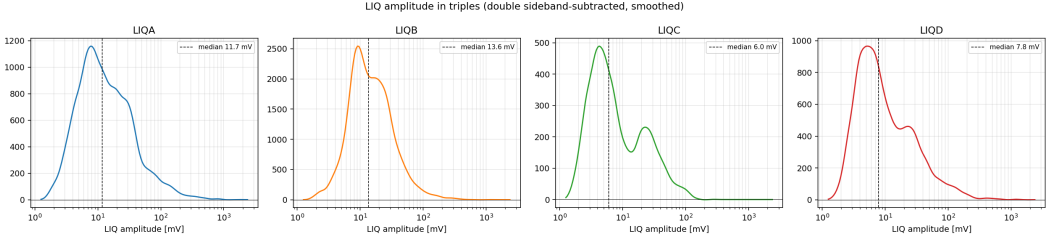


∼2–6 k net triples per PMT per step after double sideband subtraction — individually noisy; the gain-aligned stack is the evidence plot, these show the raw marching with HV.

Backup: wall spectrum unchanged by the LIQ tag



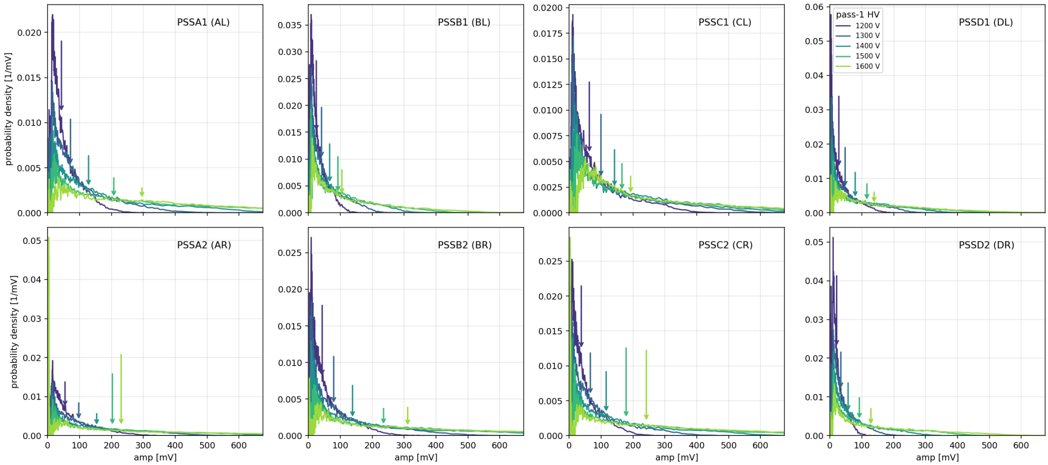
Backup: triple-tagged LIQ spectra (C/D low)



Inclusive LIQ medians: A 7.5, B 10.4, C 4.7, D 5.3 mV.

Backup: coincident spectra per step (linear mV)

run224489: wall-tagged coincident plastic spectra vs HV, linear amplitude axis (pass 1, gating OFF); arrows mark each spectrum's median = the equalization observable



Backup: trigger threshold scan at equalized gains